



13.6.5 Wrap-Up: Cool Down

1. Identify **two** strategies you can use to classify triangles using coordinate geometry.



Unit 13, Lesson 7: Centroids of Triangles



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to identify the centroid of a triangle.



13.7.1 Warm-Up: Human Hamster Wheel

A human hamster wheel is shown.

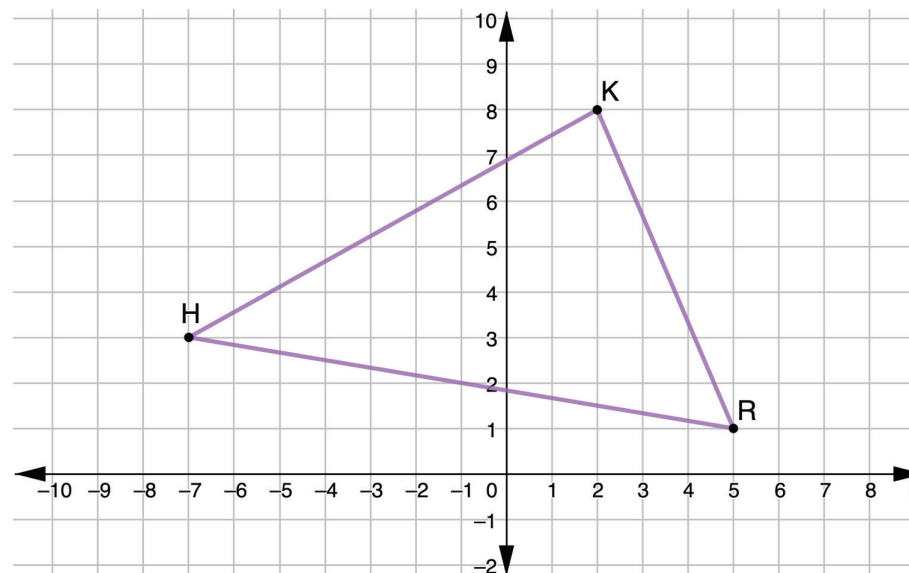


1. What is the first question that comes to your mind?
2. What are two things you notice about the human hamster wheel?



13.7.2 Exploration: Centroids of Triangles

1. $\triangle HKR$ has vertices at $H(-7,3)$, $K(2,8)$, and $R(5,1)$.



- a. Complete the table by finding the midpoint of each line segment.

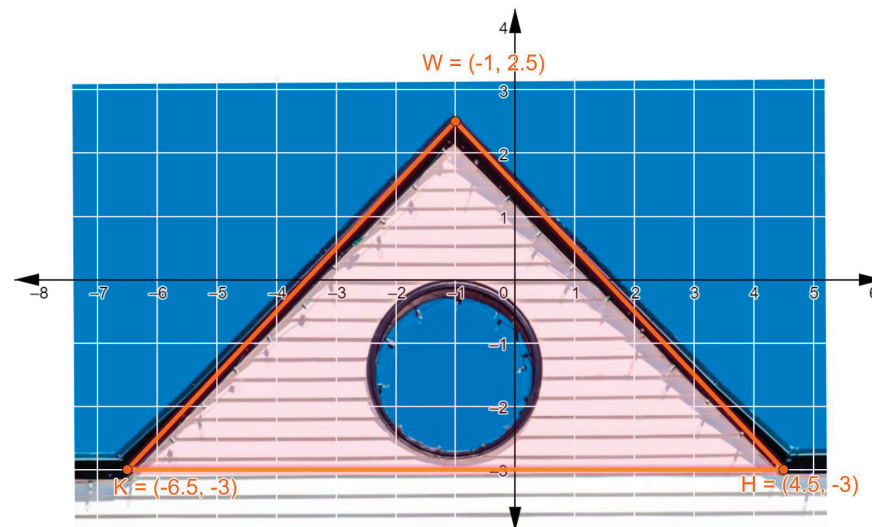
Line Segment	Work	Midpoint
$\overline{HK}$		
$\overline{KR}$		
$\overline{RH}$		

- b. Complete the steps to find the centroid of triangle HKR .

1	Graph the midpoint of line $\overline{HK}$ and label it W .
2	Using a straightedge, draw the median from W to the opposite vertex, $\angle HRK$.
3	Graph the midpoint of line $\overline{KR}$ and label it S .
4	Using a straightedge, draw the median from S to the opposite vertex, $\angle KHR$.
5	Graph the midpoint of line $\overline{RH}$ and label it M .
6	Using a straightedge, draw the median from M to the opposite vertex, $\angle HKR$.
7	Label the centroid of triangle HKR point P .

- c. What are the coordinates of the centroid of $\triangle HKR$?

2. The façade of a house is shown on the coordinate grid. The façade is represented by $\triangle WHK$ with vertices $W(-1, 2.5)$, $H(4.5, -3)$, and $K(-6.5, -3)$.



Work with your partner to estimate the location of the centroid of the triangular façade.

3. Use the centroids for $\triangle HKR$ and $\triangle WHK$ to complete the table.

Triangle	Centroid	Average of the x -coordinates of the Vertices	Average of the y -coordinates of the Vertices
$\triangle HKR$			
$\triangle WHK$			

4. Work with your partner to make a conjecture about the relationships of the centroid and the average of the x and y coordinates.

5. Ask two classmates for their conjectures. Record their conjectures and write your summary of their conjectures. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

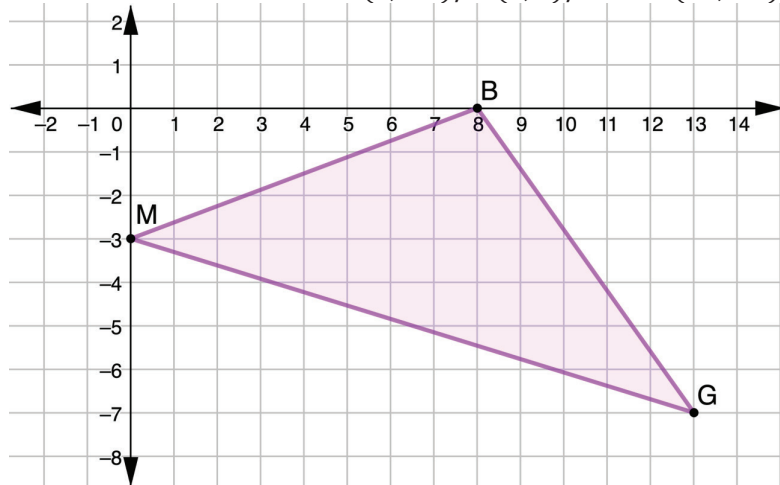
Conjecture	My Summary of Their Conjecture	Initials

6. Review your conjecture with your partner. If necessary, make any changes.



13.7.3 Collaboration: Centroids of Triangles

1. $\triangle MBG$ has vertices at $M(0, -3)$, $B(8, 0)$, and $G(13, -7)$.

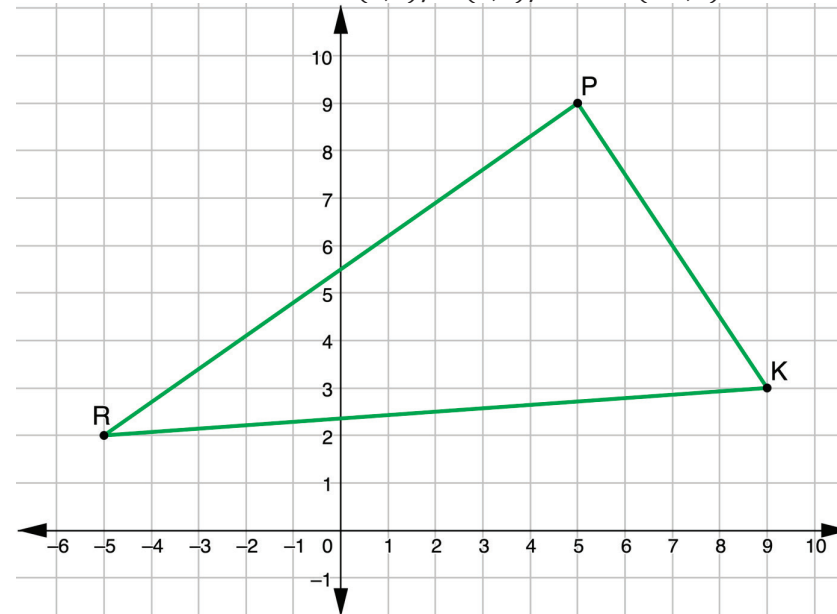


- What is the average of the x -coordinates?
- What is the average of the y -coordinates?
- What are the coordinates of the centroid?
- Verify the coordinates of the centroid by drawing the median of each side.



13.7.4 Your Turn!

1. $\triangle PKR$ has vertices at $P(5, 9)$, $K(9, 3)$, and $R(-5, 2)$.



What are the coordinates of the centroid of $\triangle PKR$?

2. $\triangle NHT$ has vertices at $N(-8, -3)$ and $H(3, -1)$ and a centroid at $(-2, -4)$.

What are the coordinates of vertex T ?

3. Artemio is finding the centroid of $\triangle WYV$ with vertices at $W(-8, 5)$, $Y(7, 8)$, and $V(-2, -3)$.

His work is shown.

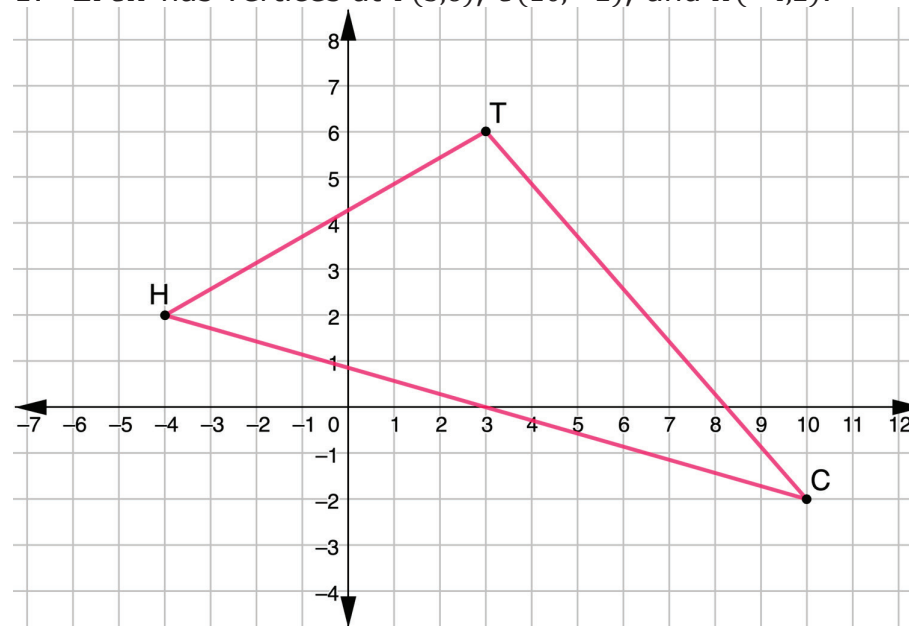
Artemio's Work	
$x = \frac{8 + 7 + 2}{3}$	$y = \frac{5 + 8 + 3}{3}$
$x = \frac{17}{3}$	$y = \frac{16}{3}$
Centroid $\left(\frac{17}{3}, \frac{16}{3}\right)$	

- Write a note to Artemio explaining his mistake.
- What are the coordinates of the centroid of $\triangle WYV$?



13.7.5 Wrap-Up: Ticket Out the Door

1. $\triangle TCH$ has vertices at $T(3, 6)$, $C(10, -2)$, and $H(-4, 2)$.



What are the coordinates of the centroid of $\triangle TCH$?